

Supplementary Material for MolBridge: Atom-Level Joint Graph Refinement for Robust Drug-Drug Interaction Event Prediction

Anonymous Submission

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1 Atom Feature Encoding Details

This section provides the complete atom feature schema used for molecular graph representation in MolBridge. Each atom in a molecule is encoded as a 75-dimensional feature vector composed of the following descriptors:

Table 1: Detailed Atom Feature Encoding Scheme

Feature	Dim.	Description
Atomic symbol	44	One-hot encoding over [C, N, O, S, F, Si, P, Cl, Br, Mg, Na, Ca, Fe, As, Al, I, B, V, K, Tl, Yb, Sb, Sn, Ag, Pd, Co, Se, Ti, Zn, H, Li, Ge, Cu, Au, Ni, Cd, In, Mn, Zr, Cr, Pt, Hg, Pb, Unknown]
Atomic degree	11	One-hot encoding over [0, 1, 2, 3, 4, 5, 6, 7, 8, 9, 10] representing the degree of atoms in a drug molecule
Implicit valence	7	One-hot encoding over [0, 1, 2, 3, 4, 5, 6] representing the implicit valence of atoms
Formal charge	1	Integer scalar representing the formal charge of the atom (typically ranges from -3 to +3)
Radical electrons	1	Integer scalar representing the number of free radical electrons (typically ranges from 0 to 2)
Hybridization	5	One-hot encoding over [SP, SP2, SP3, SP3D, SP3D2] representing the atomic hybridization mode
Aromaticity	1	Binary indicator (0/1) denoting whether the atom is aromatic
Total hydrogens	5	One-hot encoding over [0, 1, 2, 3, 4] representing the total number of hydrogen atoms attached
Total	75	

These features are extracted using RDKit 2024.9.6 and are concatenated to form the initial node feature matrix $F \in \mathbb{R}^{N \times 75}$ for each molecule, where N is the number of atoms. This feature representation captures both topological and chemical properties essential for modeling drug-drug interactions.

2 Complete Implementation Details

This section provides comprehensive experimental configurations and baseline comparison details.

2.1 Baseline Methods and Sources

For transductive experiments on Deng’s and Ryu’s datasets:

- **Results from literature:** DeepDDI, R-GCN, GoGNN, TrimNet-DDI, SSI-DDI, MUF-FIN, and MRCGNN results are directly obtained from [1].
- **Re-implemented methods:** DSN-DDI, CSSE-DDI, and TIGER are re-implemented based on their official configurations and open-source code.

For inductive generalization experiments on DrugBank:

- All baseline results and experimental protocols are obtained from [2], which provides a comprehensive evaluation framework for inductive DDI prediction under S1 (one seen drug, one unseen) and S2 (both drugs unseen) settings.

2.2 Experimental Setup for Transductive Experiments

Data splitting:

- Following the protocol in MRCGNN [1], we perform five-fold cross-validation on Deng’s and Ryu’s datasets.
- Each fold is further divided into training, validation, and test sets at a 7:1:2 ratio.

Evaluation metrics:

- Accuracy (Acc.): The proportion of correctly predicted drug pairs.
- Macro-F1: The harmonic mean of macro-precision and macro-recall.
- Macro-Precision (Macro-Pre.): Average precision across all DDI event classes.
- Macro-Recall (Macro-Rec.): Average recall across all DDI event classes.
- All results are averaged across five folds with standard deviations reported.

2.3 Training Configuration

Hyperparameters:

- Batch size: 512
- Learning rate: 0.005
- Random seed: 42
- Number of SCM layers: 3
- Optimizer: AdamW with weight decay 0.01
- Dropout rate: 0.1
- Maximum epochs: 500

- Early stopping: Based on validation accuracy and macro-F1 score

Model selection:

- Best-performing model is selected based on combined Accuracy and Macro-F1 performance on the validation set.
- Final reported results are obtained by averaging performance across all five folds.

2.4 Hardware and Software Environment

Hardware:

- CPU: Intel Xeon Platinum-8457C
- GPU: NVIDIA L20 (48GB memory)
- RAM: 128GB

Software:

- Operating System: Ubuntu 22.04.1 LTS
- Python: 3.9
- PyTorch: 2.4.0 (with CUDA 11.8 support)
- RDKit: 2024.9.6
- Additional libraries: NumPy, Pandas, scikit-learn 1.6.1, PyYAML, tqdm

3 Hyperparameter Sensitivity Analysis

We conducted systematic hyperparameter sensitivity analysis on Deng’s dataset to understand the impact of key hyperparameters on model performance. The analysis covers learning rate, batch size, number of attention heads, and number of SCM layers. Figure 1 presents a comprehensive comparison of these four key hyperparameters.

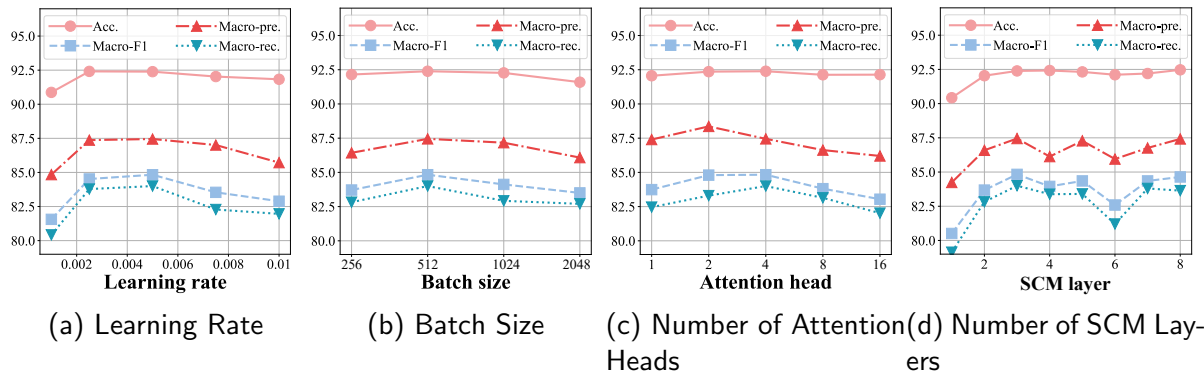


Figure 1: Comprehensive hyperparameter sensitivity analysis on Deng’s dataset. (a) Learning rate: optimal at 0.005. (b) Batch size: optimal at 512. (c) Attention heads: optimal at 4. (d) SCM layers: optimal at 3.

3.1 Key Findings

Learning Rate: We tested learning rates of 0.001, 0.0025, 0.005, 0.0075, and 0.01. As shown in Figure 1a, MolBridge achieves optimal performance across all metrics at $lr = 0.005$. Performance degrades at higher learning rates (0.0075 and 0.01), likely due to unstable optimization and overshooting. Lower learning rates (0.001 and 0.0025) also result in suboptimal performance, potentially due to slow convergence within the maximum epoch limit.

Batch Size: We examined batch sizes of 256, 512, 1024, and 2048. Results in Figure 1b indicate peak performance at batch size 512. Smaller batches (256) introduce excessive gradient estimation noise, leading to unstable training. Larger batches (1024 and 2048) compromise generalization and require greater computational resources without providing performance benefits.

Number of Attention Heads: We varied the number of attention heads in the cross-molecular interaction modeling module from 1 to 16 (specifically: 1, 2, 4, 8, 16). Figure 1c reveals that performance initially improves with more heads, peaking at 4 heads, then gradually declining. Too few heads limit the model’s capacity to capture diverse relational information, while too many heads introduce redundancy and noise in the reconstructed representations.

Number of SCM Layers: We investigated the number of Structure Consistency Module layers from 1 to 8. Performance improves as the number of layers increases from 1 to 3, but decreases beyond $n = 3$ across all metrics (Figure 1d). Insufficient layers cannot extract long-distance interaction information effectively, while excessive layers become susceptible to noise interference and over-smoothing, despite the residual connections.

3.2 Optimal Configuration Summary

Based on the sensitivity analysis, the optimal hyperparameter configuration is:

- Learning rate: 0.005
- Batch size: 512

- Attention heads: 4
- SCM layers: 3

This configuration is used for all experiments reported in the main paper.

4 Additional t-SNE Visualizations

This section provides complete t-SNE visualizations for embedding quality analysis.

4.1 High-Frequency DDI Events

Figure 2 shows t-SNE visualizations for the four most frequent DDI event types on Deng’s dataset. For each method, the upper row displays the initial pairwise feature encoding (concatenation of individual drug features), and the lower row shows the learned embedding after model training.

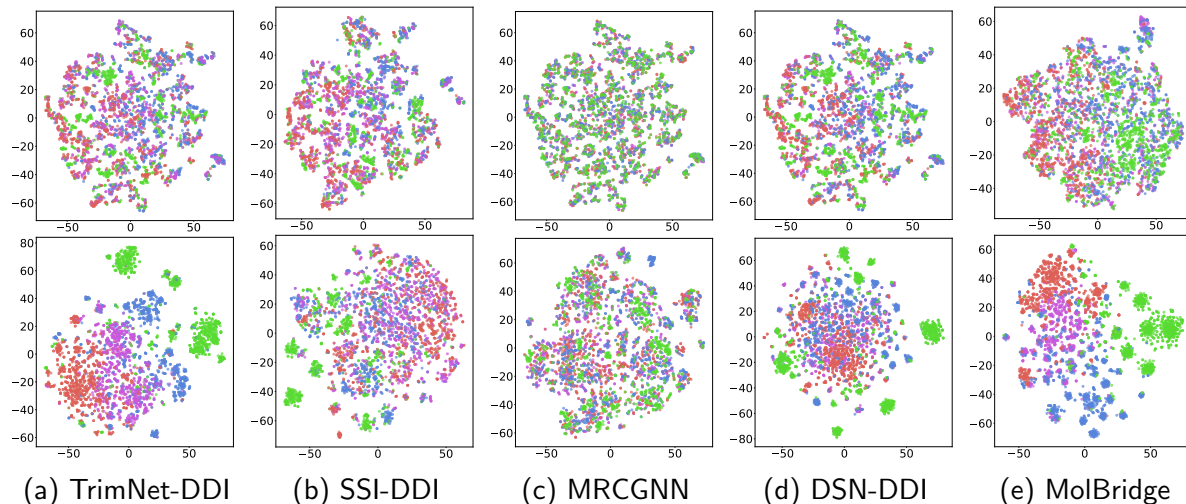


Figure 2: t-SNE visualization of the 4 most-frequent DDI events on Deng’s dataset. Upper row: initial feature encoding. Lower row: learned embeddings. MolBridge produces notably more compact and well-separated clusters compared to baseline methods.

Key observations:

- MolBridge produces notably more compact and well-separated clusters compared to baseline methods, indicating superior discriminative power.
- The transformation from initial features (upper row) to learned embeddings (lower row) is most pronounced for MolBridge, demonstrating effective feature learning.
- TrimNet-DDI and SSI-DDI show more scattered distributions, suggesting limited capability in capturing cross-molecular interactions.
- MRCGNN and DSN-DDI achieve moderate clustering quality, but still lag behind MolBridge in terms of intra-cluster compactness and inter-cluster separation.

5 Complete ATC Code Analysis for All Clusters

This section provides the complete Anatomical Therapeutic Chemical (ATC) classification analysis for representative drugs in all identified clusters (0–6). The ATC classification system is hierarchically organized with five levels:

- **Level 1** (anatomical main group): Single letter indicating the organ or system on which the drug acts.
- **Level 2** (therapeutic subgroup): Two-digit number representing the therapeutic category.
- **Level 3** (pharmacological subgroup): Single letter representing the pharmacological subgroup.
- **Level 4** (chemical subgroup): Single letter representing the chemical subgroup.
- **Level 5** (chemical substance): Two-digit number representing the specific chemical substance.

Key observations:

- **Functional coherence in clusters 3–5:** Drugs in these clusters all belong to Level 1 category “L” (Antineoplastic and immunomodulating agents), explaining their spatial proximity in the embedding space despite different specific mechanisms.
- **High similarity in cluster 4:** Both drugs (DB11901 and DB08899) share identical ATC codes through Level 4 (L02BB - Antiandrogens), with both being used for similar clinical applications (endocrine therapy for cancer treatment).
- **Therapeutic similarity in cluster 6:** DB01320 and DB00564 share the same Level 1–3 classification (N03A - Antiepileptics), despite having different chemical subgroups (hydantoin vs. carboxamide derivatives).
- **Single-drug clusters (0–3):** Each represents a distinct therapeutic category, indicating that MolBridge successfully identifies unique molecular patterns associated with specific drug classes.
- **Biological relevance without supervision:** MolBridge was trained exclusively on molecular structures without any therapeutic classification information, yet the learned representations naturally organize drugs according to their therapeutic similarities, demonstrating successful extraction of functional information from purely structural data.

Table 2: Complete ATC Code Analysis for Drugs in Each Identified Cluster

Cluster	Drug	ATC Code	Level 1	Level 2	Level 3	Level 4	Level 5
0	DB00559	C02KX01	C - Cardiovascular system	02 - Antihypertensives	K - Other antihypertensives	X - Pulmonary hypertension drugs	01 - Bosentan
1	DB04839	G03HA01	G - Genitourinary system and sex hormones	03 - Sex hormones and modulators of the genital system	H - Antandrogens	A - Antandrogens, plain	01 - Cyproterone
2	DB09280	R07AX30	R - Respiratory system	07 - Other respiratory system products	A - Other respiratory system drugs	X - Other respiratory system products	30 - Iva- cator/ Lumacaftor
3	DB00648	L01XX23	L - Antineoplastic and immunomodulating agents	01 - Antineoplastic agents	X - Other antineoplastic agents	X - Other antineoplastic agents	23 - Mitotane
4	DB11901	L02BB05	L - Antineoplastic and immunomodulating agents	02 - Endocrine therapy	B - Hormones and related agents	B - Antandrogens	05 - Enzalutamide
	DB08899	L02BB04	L - Antineoplastic and immunomodulating agents	02 - Endocrine therapy	B - Hormones and related agents	B - Antandrogens	04 - Enzalutamide
5	DB08912	L01EC02	L - Antineoplastic and immunomodulating agents	01 - Antineoplastic agents	E - Protein kinase inhibitors	C - BRAF serine/threonine kinase inhibitors	02 - Dabrafenib
6	DB01320	N03AB05	N - Nervous system	03 - Antiepileptics	A - Antiepileptics	B - Hydantoin derivatives	05 - Phenytoin
	DB00564	N03AF01	N - Nervous system	03 - Antiepileptics	A - Antiepileptics	F - Carboxamide derivatives	01 - Carbamazepine

6 Long-Distance Interaction Analysis

To systematically investigate how long-distance atomic interactions affect DDI event prediction performance, we designed a comprehensive experimental framework analyzing the distribution of average shortest path lengths between atoms within drug molecules in the Deng test set.

6.1 Distribution of Atomic Path Lengths

Figure 3 shows the density distribution of average shortest path lengths between atoms within drug molecules in the Deng test set. The vertical lines represent boundary points dividing the dataset into five equal quintiles, located at 4.16 (20%), 4.73 (40%), 5.29 (60%), and 6.01 (80%).

6.2 Performance Across Distance Groups

We conducted comparative performance analysis of MolBridge against two established baseline models (DSN-DDI and GCN-DDI) across each quintile using four standard metrics. The

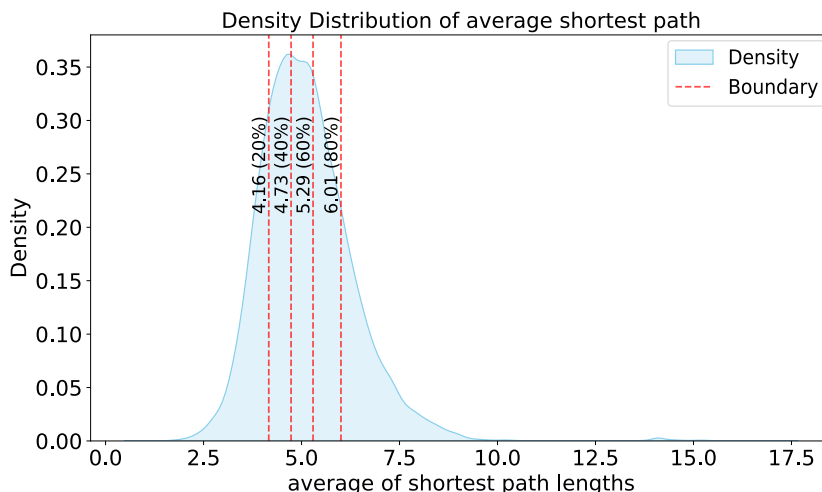


Figure 3: Density distribution of average shortest path lengths between atoms within drug molecules in the Deng test set. Vertical lines indicate quintile boundaries used for performance stratification.

complete results are shown in Figure 4.

Key findings:

- **Consistent superiority:** MolBridge significantly outperforms both baseline models across all distance segments and all metrics.
- **Distance-dependent degradation:** All models show a consistent declining performance trend as the average distance between atoms increases. In part 1 (shortest distances), MolBridge achieved 93.8% accuracy and 86.5% macro-F1 score. These metrics gradually decreased to 90.0% and 79.2% respectively in part 5 (longest distances).
- **Validation of long-range modeling:** These findings provide strong empirical evidence that long-distance atomic interactions represent an important factor constraining DDI prediction accuracy. The superior performance of MolBridge can be attributed to its Structure Consistency Module, which effectively captures and integrates features of long-distance atom interactions.

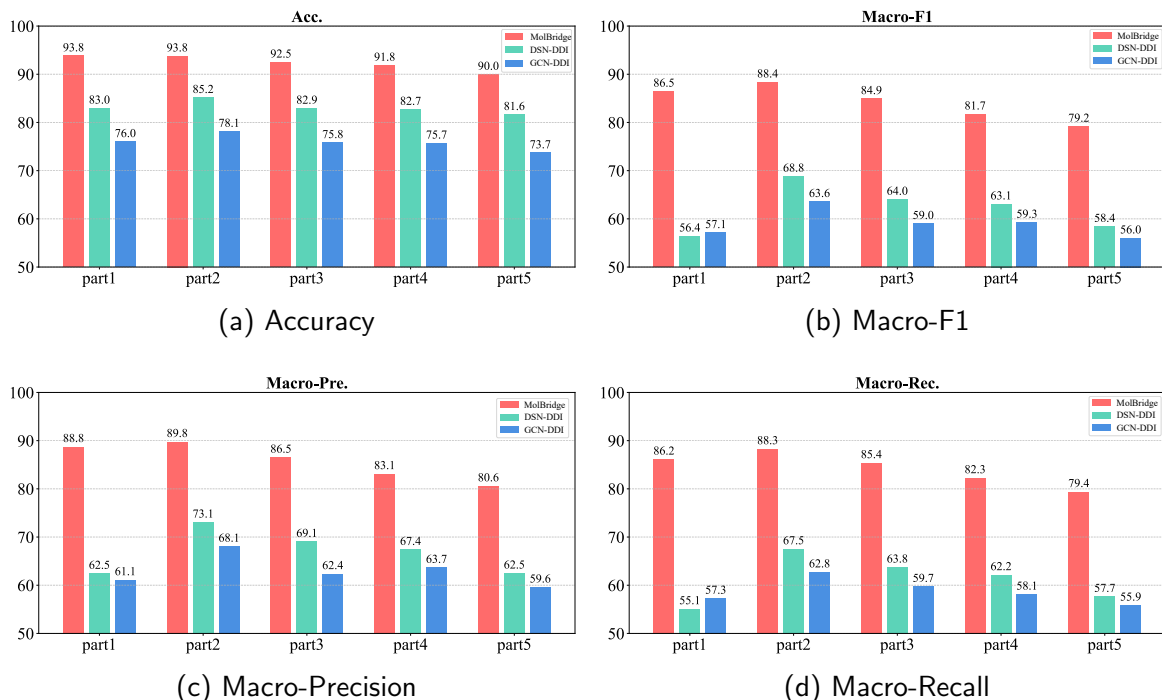


Figure 4: Complete performance comparison across different molecular distance conditions. MolBridge consistently outperforms baselines across all distance groups and maintains superior performance even for molecules with longer inter-atomic distances.

7 Summary

This supplementary material provides comprehensive details supporting the main findings presented in the MolBridge paper:

1. **Feature encoding:** Complete 75-dimensional atom feature schema used for molecular representation.
2. **Implementation details:** Comprehensive experimental configurations, baseline comparisons, training procedures, and computational environment specifications.
3. **Hyperparameter analysis:** Thorough sensitivity analysis establishing optimal configurations for learning rate (0.005), batch size (512), attention heads (4), and SCM layers (3).
4. **Visualization analysis:** Additional t-SNE visualizations for high-frequency DDI events confirming superior clustering quality and discriminative power.
5. **ATC code analysis:** Complete therapeutic classification analysis for all clusters (0–6) revealing biologically meaningful organization of learned representations.
6. **Long-distance interactions:** Systematic evaluation with complete four-metric analysis confirming the importance of modeling long-range atomic dependencies for DDI prediction.

These comprehensive analyses collectively demonstrate that MolBridge’s atom-level joint graph refinement framework effectively captures both local and global molecular interaction patterns, yielding robust, interpretable, and mechanistically relevant DDI predictions.

References

References

- [1] Zhankun Xiong, Shichao Liu, Feng Huang, Ziyang Wang, Xuan Liu, Zhongfei Zhang, and Wen Zhang. Multi-relational contrastive learning graph neural network for drug-drug interaction event prediction. In *Proceedings of the Association for the Advancement of Artificial Intelligence Conference on Artificial Intelligence*, volume 37, pages 5339–5347, 2023.
- [2] Yue Zhang, Qingyun Yao, Liang Yue, Xian Zhou, Xian-Sheng Wu, Aidong Zhang, Xiaohui Xie, and Xiaoxiao Sun. Emerging drug interaction prediction enabled by a flow-based graph neural network with biomedical network. *Nature Computational Science*, 3:1023–1033, 2023.